Two-Way Frequency Tables & Conditional Probability

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Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the municipal library, in a calculation for the science museum (case 4), if $P(A) = \frac{19}{20}$, what is $P(\text{not } A)$?

2 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the ceramic studio, a survey of 229 residents found that 51 are from an urban area and commuting by public transit, 31 are from an urban area and not commuting by public transit, 50 are from a rural area and commuting by public transit, and 97 are from a rural area and not commuting by public transit. What is the probability that a randomly selected resident is from an urban area?

3 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the botanical conservatory classified by membership status and preference between adult tickets and student tickets. If one visitor who is a Member is selected at random, what is the probability that the visitor prefers adult tickets?

Survey Results (botanical conservatory)

	adult ticket	student ticke	Total
Member	12	18	30
Non-Membe	15	25	40
Total	27	43	70

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4 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the municipal library classified by membership status and preference between hardcover books and paperback books. If a visitor who prefers hardcover books is selected at random, what is the probability that the visitor is a Non-Member?

Survey Results (municipal library)

	hardcover bo	paperback bo	Total
Member	14	19	33
Non-Membe	16	27	43
Total	30	46	76

5 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the community theater classified by membership status and preference between small planters and large planters. If one visitor from the entire survey is selected at random, what is the probability that the visitor is a Non-Member who prefers large planters?

Survey Results (community theater)

	small plante	large plante	Total
Member	16	20	36
Non-Membe	17	29	46
Total	33	49	82

6 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the regional bakery classified by membership status and preference between standard passes and express passes. If one visitor who is a Member is selected at random, what is the probability that the visitor prefers standard passes?

Survey Results (regional bakery)

	standard pas	express pass	Total
Member	18	21	39
Non-Membe	18	31	49
Total	36	52	88

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7 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the coastal aquarium classified by membership status and preference between oak saplings and maple saplings. If a visitor who prefers oak saplings is selected at random, what is the probability that the visitor is a Non-Member?

Survey Results (coastal aquarium)

	oak sapling	maple sapling	Total
Member	20	22	42
Non-Member	19	33	52
Total	39	55	94

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the meteorological station, in a group of 50 students, 30 are taking Spanish. What is the probability that a randomly chosen student is NOT taking Spanish?

9 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the urban bike-share station, in a calculation for the public transit depot (case 32), if $P(E) = 0.82$, what is $P(E')$?

10 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the sports complex, a student has 4 blue pens and 1 red pen. If one pen is picked at random, what is the probability it is red?

11 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the wind energy cooperative, a two-way table of $N = 102$ observations on science track records cell counts $p_1 = 34$ for (Section A, Physics) and $q_2 = 24$ for (Section B, Biology). What is the diagonal sum $t = p_1 + q_2$?

12 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the astronomical society, a survey of 210 residents found that 130 are from an urban area and 80 are from a rural area. Of those who are from an urban area, 114 are also commuting by public transit. Of those who are from a rural area, 58 are also commuting by public transit. How many residents in total are commuting by public transit?

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13 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the film archive, a survey of 99 patients found that 31 are receiving the treatment and 68 are receiving the placebo. Of those who are receiving the treatment, 24 are also showing improvement. Of those who are receiving the placebo, 20 are also showing improvement. How many patients in total are showing improvement?

14 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the mountain observatory, two events A and B are independent. If $P(A) = 0.4$ and $P(B) = 0.7$, what is $P(A \text{ and } B)$?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the maple syrup farm, if $P(A) = 0.4$, $P(B) = 0.5$, and A and B are independent, what is $P(A \text{ or } B)$?

16 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the planetarium, a bag has 4 red, 6 blue, and 2 green marbles. One marble is drawn, observed, and REPLACED, then a second is drawn. What is the probability that both are blue?

17 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the regional airport shuttle, a test consists of 3 true/false questions. If a student guesses on all 3, what is the probability of getting all 3 correct?

18 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the wind energy cooperative, if $P(A \mid B) = 0.6$ and $P(B) = 0.5$, what is $P(A \text{ and } B)$?

19 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the marine research vessel, two standard dice are rolled. What is the probability that the sum of the numbers is 7?

20 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the botanical conservatory, a group of 30 workshop participants includes 14 who selected adult tickets and 16 who selected student tickets. If one participant from this group is selected at random, what is the probability that the participant selected adult tickets?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the mountain observatory classified by membership status and preference between glass jars and ceramic mugs. If one visitor from the entire survey is selected at random, what is the probability that the visitor is a Non-Member who prefers ceramic mugs?

Survey Results (mountain observatory)

	glass jars	ceramic mug	Total
Member	22	23	45
Non-Member	20	35	55
Total	42	58	100

a. $\frac{35}{55}$

b. $\frac{35}{58}$

c. $\frac{23}{100}$

d. $\frac{35}{100}$

22 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the science museum classified by membership status and preference between morning shuttles and afternoon shuttles. If one visitor who is a Member is selected at random, what is the probability that the visitor prefers morning shuttles?

Survey Results (science museum)

	morning shuttle	afternoon shuttle	Total
Member	24	24	48
Non-Member	21	37	58
Total	45	61	106

a. $\frac{24}{106}$

b. $\frac{48}{106}$

c. $\frac{24}{48}$

d. $\frac{24}{45}$

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23 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the urban bike-share station classified by membership status and preference between solar panels and battery modules. If a visitor who prefers solar panels is selected at random, what is the probability that the visitor is a Non-Member?

Survey Results (urban bike-share station)

	solar panel	battery modu	Total
Member	26	25	51
Non-Member	22	39	61
Total	48	64	112

a. $\frac{26}{48}$

b. $\frac{22}{48}$

c. $\frac{22}{61}$

d. $\frac{22}{112}$

24 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the solar microgrid facility classified by membership status and preference between cotton shirts and linen shirts. If one visitor from the entire survey is selected at random, what is the probability that the visitor is a Non-Member who prefers linen shirts?

Survey Results (solar microgrid facility)

	cotton shirt	linen shirts	Total
Member	28	26	54
Non-Member	23	41	64
Total	51	67	118

a. $\frac{41}{118}$

b. $\frac{41}{64}$

c. $\frac{41}{67}$

d. $\frac{26}{118}$

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25 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the ceramic studio classified by membership status and preference between 2-pound bags and 5-pound bags. If one visitor who is a Member is selected at random, what is the probability that the visitor prefers 2-pound bags?

Survey Results (ceramic studio)

	2-pound bag	5-pound bag	Total
Member	30	27	57
Non-Member	24	43	67
Total	54	70	124

a. $\frac{30}{54}$

b. $\frac{30}{124}$

c. $\frac{57}{124}$

d. $\frac{30}{57}$

26 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the agricultural cooperative classified by membership status and preference between bench seats and floor seats. If a visitor who prefers bench seats is selected at random, what is the probability that the visitor is a Non-Member?

Survey Results (agricultural cooperative)

	bench seats	floor seats	Total
Member	32	28	60
Non-Member	25	45	70
Total	57	73	130

a. $\frac{25}{130}$

b. $\frac{32}{57}$

c. $\frac{25}{57}$

d. $\frac{25}{70}$

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27 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the publishing house classified by membership status and preference between weekday rentals and weekend rentals. If one visitor from the entire survey is selected at random, what is the probability that the visitor is a Non-Member who prefers weekend rentals?

Survey Results (publishing house)

	weekday rental	weekend rental	Total
Member	34	29	63
Non-Member	26	47	73
Total	60	76	136

a. $\frac{29}{136}$

b. $\frac{47}{136}$

c. $\frac{47}{73}$

d. $\frac{47}{76}$

28 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the public transit depot classified by membership status and preference between blueberry muffins and cinnamon scones. If one visitor who is a Member is selected at random, what is the probability that the visitor prefers blueberry muffins?

Survey Results (public transit depot)

	blueberry muffin	cinnamon scone	Total
Member	36	30	66
Non-Member	27	49	76
Total	63	79	142

a. $\frac{36}{66}$

b. $\frac{36}{63}$

c. $\frac{36}{142}$

d. $\frac{66}{142}$

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

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29 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the art gallery, a standard 6-sided die is rolled. What is the probability of rolling a number greater than 4?

a. $\frac{1}{6}$

b. $\frac{1}{3}$

c. $\frac{1}{2}$

d. $\frac{2}{3}$

30 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the robotics club, a bag contains 5 red, 4 blue, and 6 green marbles. If one marble is drawn at random, what is the probability that it is blue?

a. $\frac{4}{15}$

b. $\frac{1}{3}$

c. $\frac{2}{5}$

d. $\frac{1}{4}$

31 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the culinary institute, if the probability of rain tomorrow is 0.35, what is the probability that it will NOT rain tomorrow?

a. 0.35

b. 0.55

c. 0.65

d. 0.75

32 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the water treatment plant, a standard deck has 52 cards. What is the probability of drawing an Ace at random?

a. $\frac{1}{52}$

b. $\frac{1}{26}$

c. $\frac{1}{13}$

d. $\frac{1}{4}$

33 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the sports complex, a spinner has 8 equal sections numbered 1 through 8. What is the probability of spinning an even number?

a. $\frac{1}{4}$

b. $\frac{3}{8}$

c. $\frac{1}{2}$

d. $\frac{5}{8}$

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34 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the maple syrup farm, a jar contains 12 red candies and 18 green candies. What is the probability of randomly choosing a red candy?

a. $\frac{2}{5}$

b. $\frac{3}{5}$

c. $\frac{1}{2}$

d. $\frac{2}{3}$

35 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the planetarium, a weather station predicts that the probability of snow tomorrow is $\frac{3}{8}$. What is the probability that it will NOT snow tomorrow?

a. $\frac{1}{8}$

b. $\frac{3}{8}$

c. $\frac{5}{8}$

d. $\frac{7}{8}$

36 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the regional airport shuttle, a box contains 20 pens: 8 black, 7 blue, and 5 red. One pen is picked at random. What is the probability that it is NOT red?

a. $\frac{1}{4}$

b. $\frac{3}{4}$

c. $\frac{2}{5}$

d. $\frac{3}{5}$

37 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the wind energy cooperative, in a calculation for the community orchard (case 19), a coin is flipped once. What is the probability of landing on tails?

a. 0

b. 0.25

c. 0.50

d. 1.0

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the municipal library, a group of 33 workshop participants includes 16 who selected hardcover books and 17 who selected paperback books. If one participant from this group is selected at random, what is the probability that the participant selected hardcover books?

a. $\frac{17}{33}$

b. $\frac{16}{17}$

c. $\frac{17}{16}$

d. $\frac{16}{33}$

39 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the community theater, a group of 36 workshop participants includes 18 who selected small planters and 18 who selected large planters. If one participant from this group is selected at random, what is the probability that the participant selected small planters?

a. 1839

b. 1842

c. $\frac{18}{36}$

d. $\frac{18}{18}$

40 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the regional bakery, a group of 39 workshop participants includes 20 who selected standard passes and 19 who selected express passes. If one participant from this group is selected at random, what is the probability that the participant selected standard passes?

a. $\frac{19}{20}$

b. $\frac{20}{39}$

c. $\frac{19}{39}$

d. $\frac{20}{19}$

41 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the coastal aquarium, a group of 42 workshop participants includes 22 who selected oak saplings and 20 who selected maple saplings. If one participant from this group is selected at random, what is the probability that the participant selected oak saplings?

a. $\frac{22}{42}$

b. $\frac{20}{42}$

c. $\frac{22}{20}$

d. $\frac{20}{22}$

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42 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the mountain observatory, a group of 45 workshop participants includes 24 who selected glass jars and 21 who selected ceramic mugs. If one participant from this group is selected at random, what is the probability that the participant selected glass jars?

a. $\frac{21}{45}$

b. $\frac{24}{21}$

c. $\frac{21}{24}$

d. $\frac{24}{45}$

43 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the science museum, a group of 48 workshop participants includes 26 who selected morning shuttles and 22 who selected afternoon shuttles. If one participant from this group is selected at random, what is the probability that the participant selected morning shuttles?

a. $\frac{26}{22}$

b. $\frac{22}{26}$

c. $\frac{26}{48}$

d. $\frac{22}{48}$

44 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the urban bike-share station, a group of 51 workshop participants includes 28 who selected solar panels and 23 who selected battery modules. If one participant from this group is selected at random, what is the probability that the participant selected solar panels?

a. $\frac{23}{28}$

b. $\frac{28}{51}$

c. $\frac{23}{51}$

d. $\frac{28}{23}$

45 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the solar microgrid facility, a group of 54 workshop participants includes 30 who selected cotton shirts and 24 who selected linen shirts. If one participant from this group is selected at random, what is the probability that the participant selected cotton shirts?

a. $\frac{30}{54}$

b. $\frac{24}{54}$

c. $\frac{30}{24}$

d. $\frac{24}{30}$

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